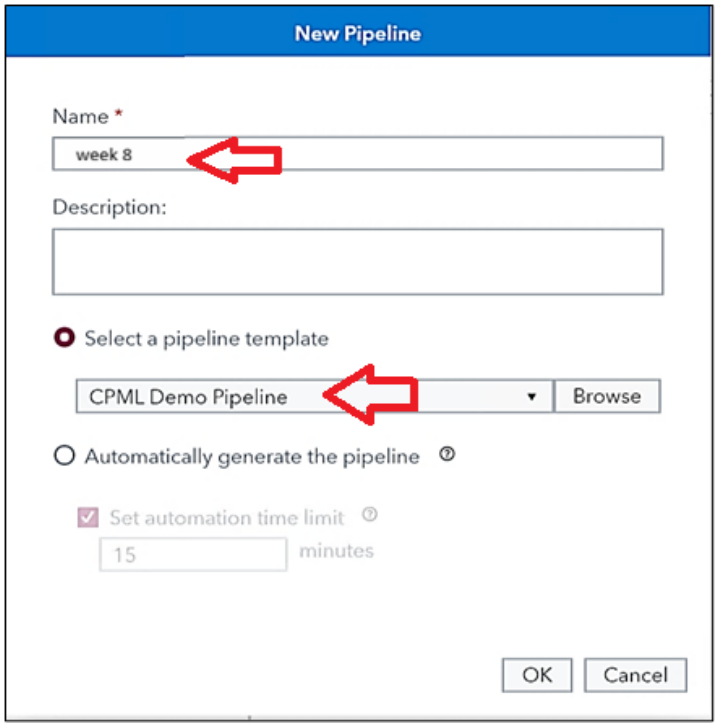
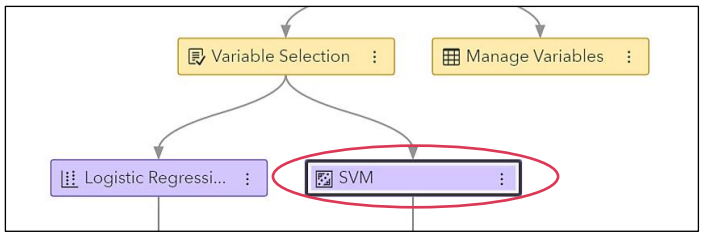
Building a Support Vector Machine Based on Default Settings

In this demonstration, you create a new pipeline using the CPML Demo Pipeline and add a Support Vector Machine node. Then, you build the support vector machine model using the node's default settings.

1. Click the plus sign (**+**) next to the pipeline tab to add a new pipeline.
2. In the New Pipeline window, enter **Week 8** in the **Name** field, access the menu under the **Select a pipeline template** property, and select **CPML Demo Pipeline**.



3. Click **OK**.
4. In the Lesson 5 pipeline, right-click the **Variable Selection** node and select **Add child node** ⇨ **Supervised Learning** ⇨ **SVM**.



5. Keep all properties for the support vector machine at their defaults.
6. Run the **SVM** node.
7. Open the results for the support vector machine model.

There are several charts and plots to help you evaluate the model's performance. The first table is Fit Statistics, which presents the support vector machine's performance considering several assessment measures.

Fit Statistics

Statistic	Training	Validation
Accuracy	0.9169	0.9154
Error	0.0831	0.0846
Sensitivity	0.3846	0.3847
Specificity	0.9905	0.9887

The *Training Results table* shows the parameters for the final support vector machine model, such as the support vectors and the margin, among others.

Training Results

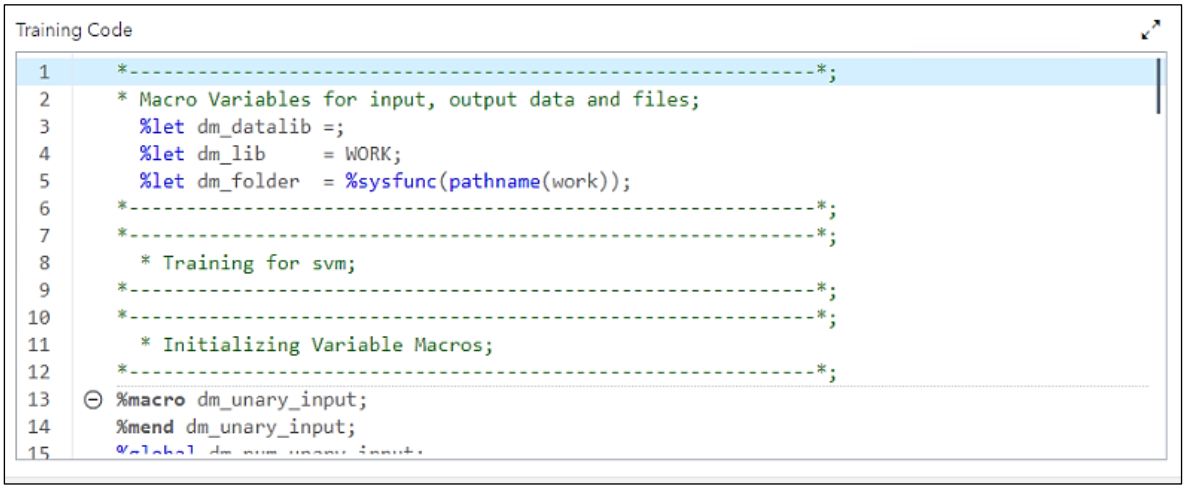
Statistic	Description	Value
WW	Inner Product of Weights	43.6941
Beta	Bias	0.0672
TotalSlack	Total Slack (Constraint Violations)	8,794.5304
LongVector	Norm of Longest Vector	2.8267
nSupport	Number of Support Vectors	39,590
nSupportInM	Number of Support Vectors on Margin	0
MaximumF	Maximum F	3.2365
MinimumF	Minimum F	-3.7650

The *Path EP Score Code window* shows the final score code that can be deployed in production.

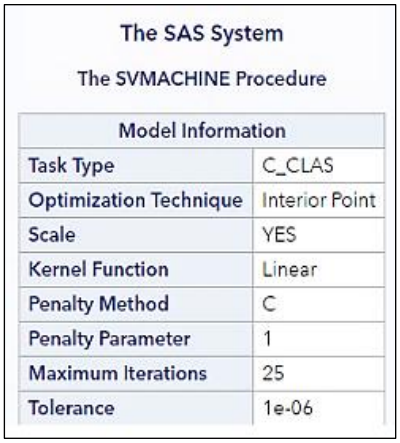
Path EP Score Code

```
1  data sasep.out;
2      dcl double "REP_BILL_DATA_USG_M03" having label n'Replacement: 3M Avg Billed Data Usag
3      dcl double "REP_BILL_DATA_USG_M06" having label n'Replacement: 6M Avg Billed Data Usag
4      dcl double "REP_CALLS_IN_OFFPK" having label n'Replacement: Calls Incoming Off-Peak' f
5      dcl double "REP_CALLS_IN_PK" having label n'Replacement: Calls Incoming Peak' format C
6      dcl double "REP_CALLS_OUT_OFFPK" having label n'Replacement: Calls Outgoing Off-Peak'
7      dcl double "REP_CALLS_OUT_PK" having label n'Replacement: Calls Outgoing Peak' format
8      dcl double "REP_DATA_DEVICE_AGE" having label n'Replacement: Avg Age of Devices on Pla
9      dcl double "REP_LIFETIME_VALUE" having label n'Replacement: Lifetime Value' format DOL
10     dcl double "REP_MB_DATA_NDIST_MO6M" having label n'Replacement: 6M Avg Billed Data Usa
11     dcl double "REP_MB_DATA_USG_M01" having label n'Replacement: MB Data Usage 1 Mth Prior
12     dcl double "REP_MB_DATA_USG_M02" having label n'Replacement: MB Data Usage 2 Mths Prio
13     dcl double "REP_MB_DATA_USG_M03" having label n'Replacement: MB Data Usage 3 Mths Prio
14     dcl double "REP_MB_DATA_USG_ROAMM01" having label n'Replacement: MB Data Usage Roam 1
15
```

Similarly, the *Training Code window* shows the train code that can be used to train the model based on different data sets or on different platforms.



Finally, the *Output window* shows the final support vector machine model parameters, the training results, the iteration history, the misclassification matrix, the fit statistics, and the predicted probability variables.

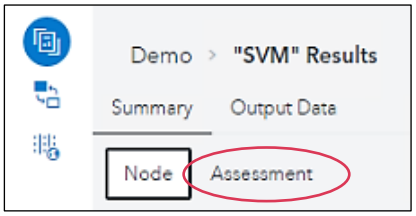


The SAS System

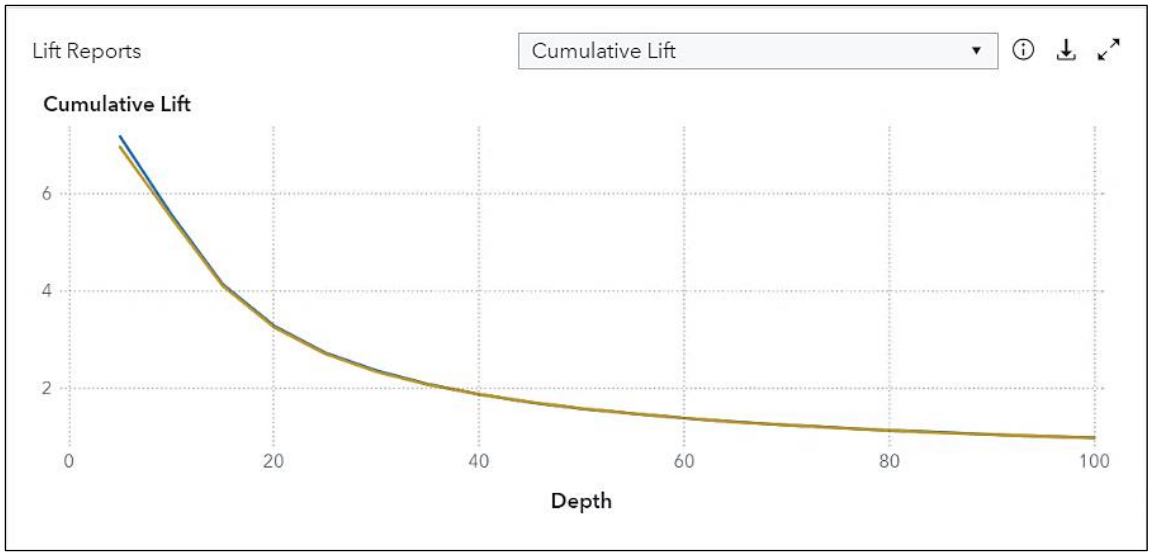
The SVMACHINE Procedure

Model Information	
Task Type	C_CLAS
Optimization Technique	Interior Point
Scale	YES
Kernel Function	Linear
Penalty Method	C
Penalty Parameter	1
Maximum Iterations	25
Tolerance	1e-06

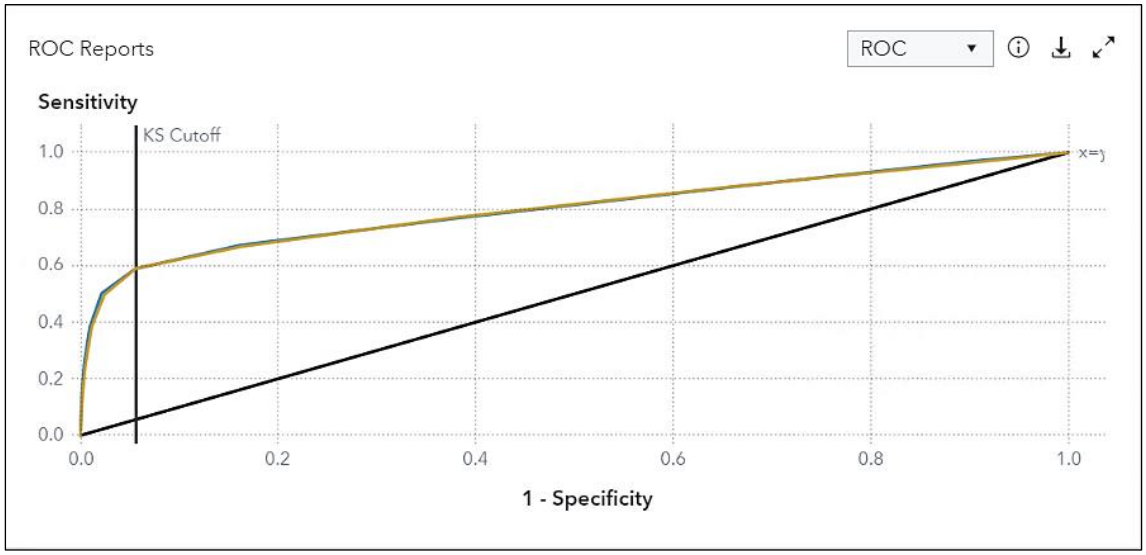
8. Click the **Assessment** tab.



The first chart is *Lift Reports*, which by default shows *Cumulative Lift*. Cumulative Lift shows the model's performance ordered by the percentage of the population. This chart is very useful for selecting the model based on a particular target of the customer base. It shows how much better the model is than the random events.



For a binary target, you also have the **ROC Reports** output, which shows the model's performance in terms of ROC curve by considering the true positive rate and the false positive rate. It is good to foresee the performance on a specific business event when all positive cases are selected. It shows that the model's performance based on the positive cases were predicted right and the positive cases were predicted wrong.



Finally, you have the Fit Statistics output, which shows the model's performance based on some assessment measures, such as average squared error.

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.1164
churn	VALIDATE	0	0	16,967	0.1169

The Fit Statistics table shows an average squared error of 0.1169 on the VALIDATE partition. Your performance metrics values can be different due to either the differences in the data preprocessing or the result of the distributed computing outputs.

9. Close the Results window.

End of Demonstration

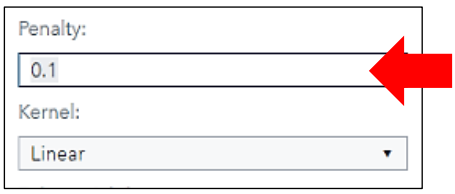
Changing the Methods of Solution for a Support Vector Machine

In this demonstration, you change the default settings of the Support Vector Machine node in the **Week 7 practical lesson** pipeline and modify its methods of solution parameters.

1. Recall that the average squared error of the previous model, based on the default settings, was 0.1169 on the VALIDATE partition.

   Change some of the default settings assigned to the solution methods to improve the support vector machine's performance.

2. Change the **Penalty** property from 1 to **0.1**.



   The Penalty (C parameter) tells the SVM optimisation how much you want to avoid misclassifying each training example. A smaller value of C causes the optimiser to look for a larger-margin separating hyperplane, even if that hyperplane misclassifies more points. A larger penalty value usually provides a more robust model. However, it might overfit.

3. Run the **Support Vector Machine** node.
4. Open the results for the node.
5. Click the **Assessment** tab. Examine the Fit Statistics window.

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.0964
churn	VALIDATE	0	0	16,967	0.0970

6. The average squared error for the tuned support vector machine model is 0.0970 on the VALIDATE partition. This fit statistic is better (smaller) than the first model, which used the default settings.
7. Close the Results window.

End of Demonstration

Changing the Kernel Function for a Support Vector Machine

In this demonstration, you change the previous settings of the Support Vector Machine node in the **Week 7 practical lesson** pipeline. You modify the kernel function and other parameters and compare this model's performance to the other model in the pipeline.

1. Recall that the previous model, based on changing the penalty term, achieved an average squared error of 0.0970 on the VALIDATE partition. Changing only the penalty term led to a model that was an improvement (at least based on ASE) compared to the SVM model built under the defaults.

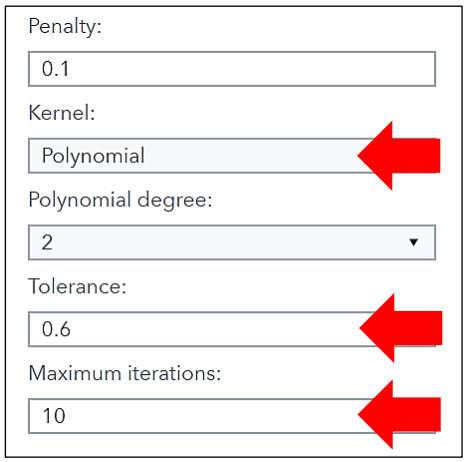
   Try to improve the support vector machine performance by changing some of the default settings assigned to the kernel function parameters.

2. For **Kernel**, change the function from Linear to **Polynomial**. Leave **Polynomial degree** as **2**.

   We are just trying to make the hyperplane decision boundary between the classes different by changing the kernel. Linear kernels usually work fine for linearly separable data. However, if the data are not linearly separable, a linear kernel does not classify well.

3. For **Tolerance**, change 0.000001 to **0.6**. Tolerance specifies the minimum number at which the iteration stops.

4. For **Maximum iterations**, change 25 to **10**. This property specifies the maximum number of iterations allowed with each try. In some cases, you can obtain a good model in fewer than five iterations.



5. Run the **Support Vector Machine** node.

6. Open the results for the node.

7. Click the **Assessment** tab. Examine the Fit Statistics window.

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.0926
churn	VALIDATE	0	0	16,967	0.0937

   The average squared error for the tuned support vector machine model is 0.0937 on the VALIDATE partition. This fit statistic is slightly better than the previous model.

8. Close the Results window.

Some improvement in the model performance has been observed.

End of Demonstration